



Department of Computer Science and Engineering
Jashore University of Science and Technology
Semester Final Examination-2020
1st Year 2nd Semester, Session: 2019-20

Course Code: CSE 1201
Course Title: Data Structure

Time: 3 00 hours
Marks: 72

[N.B. Answer any 06 sets questions from following 08 sets questions.
The figures in the right margin indicate full marks.]

- ✓ a) Discuss the differences between linear and non-linear data structure. (03)
- ✓ b) Suppose a five dimensional array A is declared using A(2:8, -4:1, 6:10, 3:7, -2:6). Consider the element A[3, -1, 8, 4, 2] in A. Find the effective indices E₁, E₂, E₃, E₄, E₅ and the address of the element in row-major order and column-major order, assuming Base(A) = 400 and there are w = 4 words per memory location. (05)
- c) Translate the infix notation $A * (B + D) / E - F * (G + H / K)$ to its equivalent postfix notation using stack. (04)
- ✓ a) How to minimize stack overflow? Explain with example(s). (02)
- b) Consider the priority queue in the following figure which is maintained as a one way list. (04)

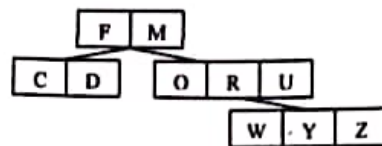
	INFO	PRN	LINK
1	A	2	6
2			7
3	B	4	4
4	D	4	9
5	E	1	1
6	G	2	3
7			10
8	C	5	0
9	F	4	8
10			11
11			12
12			0

AVAIL → 2

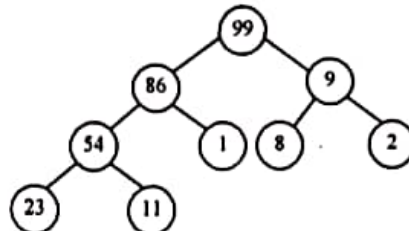
START → 5

- i) Describe the structure after (X, 7), (Y, 3), (Z, 2) and (W, 1) are added to the queue.
- ii) Describe the structure if, after the preceding insertions, 3 elements are deleted.
- c) What is the running time of BFS if we represent its input graph by an adjacency matrix and modify the algorithm to handle this form of input? (03)
- d) Show the steps if you apply quick sort algorithm for the following data in ascending order: 66, 33, 40, 22, 53, 88, 60, 11, 80, and 20. (03)
- ✓ a) Consider the following deque of characters where DEQUE is a circular array which allocates six memory cells: LEFT = 2, RIGHT = 4, DEQUE: —, A, C, D, —, —. Describe the deque while the following operations take place: (05)
- G is added to the right of the deque.
 - Three letters on the right are deleted.
 - K, L, M are added to the left of the deque.
 - One letter on the left is deleted.
 - T is added to the left of the deque.
- b) Consider a hash table of size 7 with hash function $h(k) = k \bmod 7$. Draw the table that results after inserting in the given order, the values, 12, 26, 7, 23, 17 for each of the scenarios below: (03)
- When collisions are handled by separate chaining.
 - When collisions are handled by linear probing.
 - When collisions are handled by double hashing.
- Using a second hash function $h'(k) = 5 - (k \bmod 5)$.
- c) How binary search algorithm works? Explain with suitable example(s). Also discuss the limitation of this algorithm. (04)
- ✓ a) Discuss the steps of merge sort using the following list as example 44, 33, 11, 55, 77, 44, 40, 90 and 60. (03)
- b) Why garbage collection is important? (04)
- c) Compress the sentence "Today is Sunday" using Huffman coding scheme and also show the effectiveness of the compression. (05)
- ✓ a) Using the definition of Ackerman functions find the value of A(1, 4) and show the each call and return values. (05)

- b) The following figure is represented as a 4-way search tree. Draw the final tree after deletion of R and F. (04)

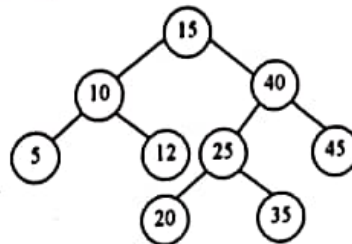


- c) What are the difference between splay tree and binary tree? Explain with example. (03)
6. a) What are the differences between BST and heap? When to use a heap and when to use a BST? (03)
- b) Consider the following heap: (07)



For each of the following questions use the original heap. Do not use the heap obtained after each answer.

- Represent the heap as an array indexed by 1 through 9.
 - Delete the element 86 from the heap shown in the above figure.
 - Delete the element 2 from the heap shown in the above figure.
 - Delete the element 8 from the heap shown in the above figure.
 - Is it max heap or min heap? Sorting the elements in descending order by heap sort.
- c) Define full binary tree and complete binary tree. Convert the expression $2*3/(2-1)+5*(4-1)$ into binary tree. (02)
7. a) Consider the following AVL tree: (04)



- Insert the key 41 into the tree and rebalance if needed. Draw the final and all intermediate trees that you need.
 - Remove the key 10 from the original tree and rebalance if needed. Draw the final tree and all intermediate trees that you need.
- b) Discuss the time complexity of bubble sort. (03)
- c) Suppose a data set S contains n elements. Compare the time complexity T[1] of Linear search algorithm with the time complexity T[2] of the binary search algorithm when (i) n=100 and (ii) n=1000. (05)
8. a) Write down the recursive formula of the "Tower of Hanoi" problem. (02)
- b) Write down the $O(n+k)$ complexity algorithm for sorting an array. (03)
- c) Define Stack and Queue. Given the following numbers 10, 4, 6, 8, 9 and 3, write (03)
- What would be the value to the stack, if three times push, one time pop, one time push and one time pop. Where push and pop operation refer to insert and pull an element to the stack respectively.
 - What would be the value to the queue, if two times enqueue, one-time dequeue, three times enqueue and four times dequeue.

Write down the operations for making empty stack to the numbers.

- Define Linked list. Implement a stack using a singly linked list L. The operations PUSH and POP should still take $O(1)$ time. (04)



Department of Computer Science and Engineering
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Semester Final Examination-2020
1st Year 2nd Semester, Session: 2019-20

Course Code: MATH 1201

Course Title: Differential Equations and Geometry

Time: 3.00 hours

Marks: 72

[N.B. Answer any 06 sets questions from following 08 sets questions.
The figures in the right margin indicate full marks.]

1. a) Define differential equation and partial differential equation with examples. 04
b) Solve the following differential equations 08
i) $\frac{dy}{dx} + \frac{y^2 + y + 1}{x^2 + x + 1} = 0$ ii) $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y$
2. a) Define homogeneous differential equation. Solve $(x^2 + y^2)dx = 2xy dy$. 04
b) Verify that the equation $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^2y - 3x^2y^2 - 5y^4)dy = 0$ is exact and solve it. 04
c) Define Bernoulli equation. Solve $(x-2)\frac{dy}{dx} = y + (x-2)^3$. 04
3. a) Define integrating factor. Solve $x\frac{dy}{dx} = y + x^3 + 3x^2 - 2x$. 06
b) Define linear and non-linear differential equation with example. Solve the differential equations $(xy^2 + x)dx + (yx^2 + y)dy = 0$. 06
4. a) What is the general form of the linear differential equation of second order? 02
b) Discuss the method for finding the complementary function. 03
c) Apply the method of variation of parameters to solve 07
i) $y_2 + n^2y = \sec x$, ii) $y_2 + 4y = \sec 2x$
5. a) Solve the initial value problem $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} - 3y = 2e^x - 10 \sin x$, $y(0) = 2$, $y'(0) = 4$ using a suitable method. 06
b) Solve $\frac{d^2y}{dx^2} + 6y = \sin 4x$. 06
6. a) Find the transformation equation of $5x^2 + 6xy + 5y^2 - 4x + 4y - 4 = 0$. 06
b) If by the rotation of the rectangular co-ordinate axes about a fixed origin in the expression $5ax^2 + 2hxy + by^2$ changes to $a'x'^2 + 2h'x'y' + b'y'^2$ then show that $a + b = a' + b'$ and $ab - h^2 = a'b' - h'^2$. 06
7. a) Prove that a homogeneous equation of the n^{th} degree represents n^{th} straight line which all pass through the origin. 06
b) Prove that the straight line represents by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will be equidistance from the origin if $f^4 - g^4 = c(bf^2 - ag^2)$. 06
8. a) Establish the relation between direction cosines and direction ratios. 02
b) Find the distance of A (1, -2, 3) from the line PQ through P (2, -3, 5) which makes equal angles with the axes. 05
c) Show that the lines $\frac{x+1}{2} = \frac{y-2}{2} = Z$ and $\frac{x-1}{6} = \frac{y+1}{1} = \frac{z-3}{5}$ are coplanar. Also find the point of intersection and equation of the plane containing them. 05



Department of Computer Science and Engineering

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Semester Final Examination-2021

1st Year 2nd Semester, Session: 2019-20

Course Code: CSE 1203

Course Title: Digital Logic Design

Time: 3 00 hours

Marks: 72

[N.B. Answer any 06 sets questions from following 08 sets questions.

The figures in the right margin indicate full marks.]

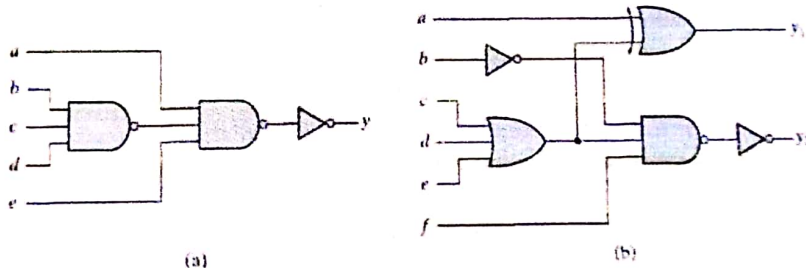
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6

1. a) i. What is the largest binary number that can be expressed with 14 bits? What are the equivalent decimal and hexadecimal numbers? 04
- ii. Perform subtraction on (2043-6152) using the 10's complement of the subtrahend. Where the result should be negative. Find its 10's complement and affix a minus sign. Verify your answer.
- b) i. Perform the following division in binary: $111011 \div 101$. 04
- ii. Find the 16's complement of B2FA.
- c) The following is a string of ASCII characters whose bit patterns have been converted into hexadecimal for compactness: 73 F4 E5 76 E5 4A EF 62 73. Of the eight bits in each pair of digits, the leftmost is a parity bit. The remaining bits are the ASCII code. 04
- i. Convert the string to bit form and decode the ASCII.
- ii. Determine the parity used: odd or even?
2. a) Implement the Boolean function 03

$$F = xy + x'y' + y'z$$

- i. With AND, OR, and inverter gates,
- ii. With NOR and inverter gates,
- iii. With NAND and inverter gates.
- b) Show that a positive logic NAND gate is a negative logic NOR gate and vice versa. 03
- c) Write Boolean expressions and construct the truth tables describing the outputs of the circuits described by the following logic diagrams: 06



3. a) Express the following function as a sum of minterms and as a product of maxterms: 02

$$F(A, B, C, D) = B'D + A'D + BD$$

- b) Simplify the following Boolean functions, using Karnaugh maps: 06

$$F(w, x, y, z) = \sum (2, 3, 12, 13, 14, 15)$$

$$F(A, B, C, D) = \sum (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

- c) Implement the following Boolean function F, together with the don't-care conditions d, using no more than two NOR gates: 04

$$F(A, B, C, D) = \sum (2, 4, 6, 10, 12)$$

$$d(A, B, C, D) = \sum (0, 8, 9, 13)$$

Assume that both the normal and complement inputs are available.

4. a) Implement the following four Boolean expression with three half adders. 06

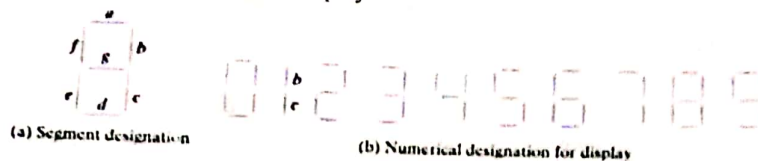
$$D = A \oplus B \oplus C$$

$$E = A'BC + AB'C$$

$$F = ABC' + (A' + B')C$$

$$G = ABC$$

- b) An ABCD to seven segment decoder is a combinational circuit that converts a decimal digit in BCD to an appropriate code for the selection of segments in an indicator used to display the decimal digit in a familiar form. The seven outputs of the decoder (a, b, c, d, e, f, g) select the corresponding segments in the display, as shown in the following fig(a). The numeric display chosen to represent the decimal digit is shown in fig(b). Using a truth table and Karnaugh maps, design the BCD to seven segment decoders, using a minimum number of gates. The six invalid combinations should result in a blank display.



5. a) Explain how a S-R FlipFlop store a bit. 03
 b) What is the limitation of S-R FlipFlop? Explain how the limitation can be resolved. 04
 c) Define Race Around condition in J-K FlipFlop. How can you overcome the problem? Explain with an appropriate figure and waveforms. 05
6. a) Implement a full adder with two 4 x 1 multiplexers. 03
 b) Implement the following Boolean function with a multiplexer. 03

$$F(A, B, C, D) = \sum (0, 2, 5, 7, 11, 14)$$

 c) A sequential circuit has JK flip-flops A and B and one input x. The circuit is described by the following flip-flop input equations: 06

$$J_A = x \quad K_A = B'$$

$$J_B = x \quad K_B = A$$

 i) Derive the state equations $A(t+1)$ and $B(t+1)$ by substituting the input equations for the J and K variables.
 ii) Draw the state diagram of the circuit.
7. a) Show that the dual of the exclusive-OR is also its complement. 03
 b) Implement the following Boolean expression with exclusive-OR and AND gates: 03

$$F = AB'CD' + A'BCD' + AB'C'D + A'BC'D$$

 c) Write the Boolean equations and draw the logic diagram of the circuit whose outputs are defined by the following truth table: 06

P.T.O

f_1	f_2	a	b	c
1	0	0	0	0
0	0	0	0	1
0	1	0	1	0
1	1	0	1	1
0	1	1	0	0
0	1	1	0	1
1	1	1	1	0
1	0	1	1	1

8. a) Use 74LS138 ICs to design a 1 of 32 decoder. 04
 b) What is Multiplexer? Design an 8-input MUX and explain its operation. 04
 c) How can you use a 74138 IC as a DEMUX. 02
 d) Draw the logic diagram for the 7442 BCD to decimal decoder. 02

Department of Computer Science and Engineering
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1st Year 2nd Semester Examination-2020

Course Title: Physics for Computer Engineers

Course Code: PHY 1201

Time: 3 hours

Full Marks: 72

N.B. There are eight questions, answer any six questions.

The figures shown in the margin indicate full marks.

1. (a) Describe different types of transmission of heat. 4
(b) Explain van der Waals equation of state. 4
(c) A cylinder contains 12 L of oxygen at 20^o C and 15 atm. The temperature is raised to 35^o C and volume is reduced to 8.5 L. What is the final pressure of the gas in atmospheres? Assume that the gas is ideal. 4
2. (a) State and explain the 1st law of thermodynamics. 3
(b) Describe Carnot cycle. 5
(c) An engine manufacturer makes the following claims: An engine's heat input per second is 9.0 KJ at 435 K. The heat output per second is 4.0 KJ at 285 K. Do you believe claims? 4
3. (a) Discuss damped oscillations and force oscillations. Give figures if necessary. 4
(b) Find amplitude and phase when two waves of the same frequency are added. 8
4. (a) What do you know about reverberation time? 2
(b) State and explain Doppler's effect. 4
(c) An wave is given by $\Psi(x, t) = \frac{3}{[10(x-vt)^2+1]}$. Show that this is a solution to the one-dimensional differential wave equation. 6
5. (a) Explain interference and diffraction 4
(b) Discuss constructive and destructive interference of two waves. 8
6. (a) Categorize ultrasonic, audible and infrasonic and supersonic waves. 4
(b) Describe the formation of beats. 3
(c) What is Coulomb's law. 3
(d) Explain transverse wave and longitudinal wave. 2
7. (a) What are Lissajou's Figure? Write down the uses of Lissajou's Figure. 3
(b) Describe the Storing Energy in a Capacitor. 5
(c) A 2000 μ F capacitor is charged through a 1 kW resistor using a 6 V supply. Calculate (i) the charging current after 2.5 second, (ii) the charge on the plates after 2.5 second. 4
8. (a) Write short notes on electric field. 2
(b) Discuss single-slit Fraunhofer diffraction. 6
(c) Deduce gas laws. 4

Department of Computer Science and Engineering
Jashore University of Science and Technology

1st Year 2nd Semester Examination-2020

Course Title: Electronic Devices and Circuits

Course Code: EEE 1201

Time: 3 hours

Full Marks: 72

N.B. There are eight questions, answer any six questions.
 The figures shown in the margin indicate full marks.

1. (a) What do you mean by polycrystalline and amorphous solid? Explain the processing method of p-type material with necessary figure. 5
- (b) Which material from Si and Ge is better, Explain. 3
- (c) Determine the dc bias voltage V_{CE} and the current I_C for the Fig. 1(c) 4

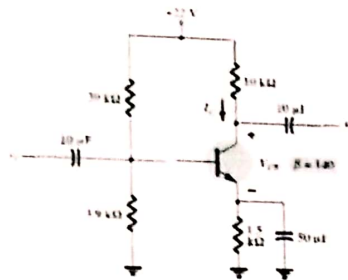


Fig. 1(c)

2. (a) What do you mean by PIV? Explain the I-V characteristics of diode. 4
- (b) Define Rectifier circuit. Explain the working principal of full wave bridge rectifier with necessary figure. 4
- (c) Determine the quiescent levels of I_{CQ} and V_{CEQ} for the Fig.2(c) 4

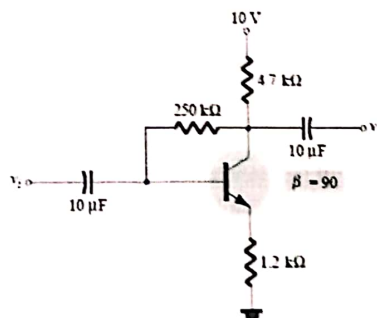


Fig. 2(c)

3. (a) What do you mean by transistor? Draw the symbol of npn and pnp transistor. 3
- (b) Define operating point. For the configuration of Fig. 3(b); Determine (i) I_B and I_C (ii) V_{CE} , V_B , V_C , V_{BC} . 5

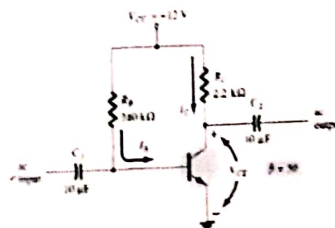


Fig. 3(b)

- (c) What do you mean voltage regulator. Design and explain a circuit which input 220 V AC and output 10 V DC. 4

4. (a) Define: Fermi Energy, Intrinsic, Extrinsic material and Doping.
 (b) What is form factor? Prove that 59.4% of input power is lost in the rectifier circuit.
 (c) For the network of Fig. 4(c); Determine (i) I_B and I_C (ii) V_{CE} , V_C , V_E , V_B and V_{BC} .

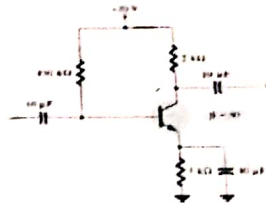


Fig. 4(c)

5. (a) What are the differences between BJT and FET? Draw the graphic symbol of different types of MOSFET. 4
 (b) Why FET is called unipolar device? Draw and explain the characteristics curve of n-channel JFET. 4
 (c) For the network of Fig. 5(c); Determine (i) V_{GS} and I_D (ii) V_{DS} , V_D , V_G , V_S . 4

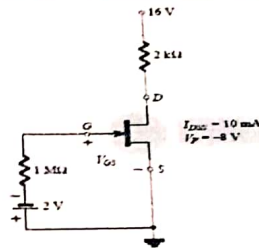


Fig.5(c)

6. (a) What are oscillators? Explain the operation of a basic oscillator. 3
 (b) Classify power electronic devices. Explain the characteristics of power electronics devices. 3
 (c) Define BJT stability factor. Determine R_C and R_B for a fixed-bias configuration if $V_{CC} = 12$ V, $V_B = 80$, and $I_{CQ} = 2.5$ mA with $V_{CEQ} = 6$ V. Use standard values. 3
 (d) Draw the DC load line of a transistor. 3
 7. (a) What do you mean by virtual ground and offset null? Explain the process of offset null adjustment with proper circuit diagram. 4
 (b) Draw with the value of resistance and explain an Op-Amp circuit which have five input and the output is the average of the inputs. 4
 (c) Design an Op-Amp circuit that will produce an output equal to $-(4V_1 + V_2 + 0.1V_3)$. Write an expression for the output and sketch its output waveform when $V_1 = 2\sin\omega t$, $V_2 = +5$ V dc and $V_3 = -100$ V dc. 4
 8. (a) What is Op-amp? What is the input and output impedance of it? What are the applications of Op-amp? 3
 (b) Derive the relation between input and output voltage [$V_{out} = V_{in} (-R_F/R_1)$] of inverting amplifier. 2
 (c) Explain briefly about op-amp integrator. 4
 (d) Why open loop is not used for linear application? Find the outputs v_{out1} , v_{out2} and v_{out3} of the following Fig. 8 (d) 3

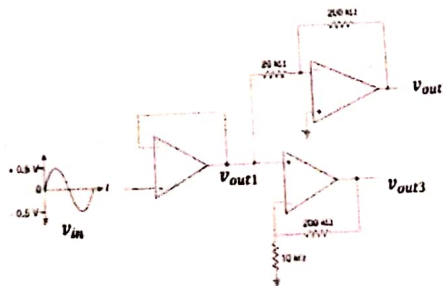


Fig. 8(d)